```
### QUERY 1 ###
================================================================================
CUSTOMER (David): What are the pricing plans available for your software?
================================================================================

    [Intent Classification] 'What are the pricing plans available for your software?' -> intent=Sales, department=Sales Support
    [Sales Support] RAG retrieved 3 chunk(s) from ['pricing_guide.txt']. High-risk=False (n/a).
    [Supervisor] Draft response validated and approved for sending.
    [Memory Save] Interaction persisted to memory.db (SQLite).

FINAL RESPONSE TO CUSTOMER:
   Thank you for contacting Sales Support. Based on our records:
- PRICING GUIDE - ABC Technologies
- SUBSCRIPTION PLANS
- 4. Enterprise Plan - Custom Pricing
   - Unlimited users
   - Unlimited storage
   - SLA-backed uptime guarantee
   - On-premise deployment option
   - Dedicated technical support team
```

```
### QUERY 2 ###
================================================================================
CUSTOMER (David): I forgot my account password.
================================================================================

    [Intent Classification] 'I forgot my account password.' -> intent=Account, department=Account Support
    [Account Support] RAG retrieved 3 chunk(s) from ['company_policy.txt', 'faq.txt']. High-risk=False (n/a).
    [Supervisor] Draft response validated and approved for sending.
    [Memory Save] Interaction persisted to memory.db (SQLite).

FINAL RESPONSE TO CUSTOMER:
   Thank you for contacting Account Support. Based on our records:
- Q: How do I reset my password?
A: Go to the login page and click "Forgot Password". Enter your registered email address and follow the link sent to your inbox to set a new password.
- Q: How do I deactivate my account temporarily?
A: Go to Account Settings > Account Status and select "Deactivate Account". You can reactivate anytime by logging back in within 90 days.
- ACCOUNT CLOSURE POLICY
Account closure requests must be verified by a human supervisor to prevent accidental or fraudulent closures.
All outstanding invoices must be settled before an account can be closed.
Closed accounts can be reactivated within 90 days by contacting support.
```

```
### QUERY 3 ###
================================================================================
CUSTOMER (David): My application crashes whenever I upload a file.
================================================================================

    [Intent Classification] 'My application crashes whenever I upload a file.' -> intent=Technical, department=Technical Support
    [Technical Support] RAG retrieved 3 chunk(s) from ['technical_manual.txt']. High-risk=False (n/a).
    [Supervisor] Draft response validated and approved for sending.
    [Memory Save] Interaction persisted to memory.db (SQLite).

FINAL RESPONSE TO CUSTOMER:
   Thank you for contacting Technical Support. Based on our records:
- APPLICATION CRASH ISSUES
If the application crashes while uploading a file, check the following:
1. Ensure the file size does not exceed the 200MB upload limit.
2. Confirm the file format is supported (CSV, XLSX, PDF, DOCX, PNG, JPG).
3. Clear the browser cache and cookies, then retry the upload.
4. Disable browser extensions that may interfere with file uploads.
5. If the issue persists, try uploading from a different browser or device.
6. Check your internet connection stability during large file uploads.
- GENERAL TROUBLESHOOTING
Restarting the application and clearing local cache resolves most minor glitches.
Persistent technical issues should be escalated with a screenshot and error log for faster resolution.
- INSTALLATION ISSUES
For desktop application installation failures:
1. Verify the operating system meets minimum requirements (Windows 10+/macOS 11+).
2. Run the installer as an administrator.
3. Temporarily disable antivirus software during installation.
4. Ensure at least 2GB of free disk space is available.
```

```
### QUERY 4 ###
================================================================================
CUSTOMER (David): I need a refund for my annual subscription.
================================================================================

   *** HUMAN-IN-THE-LOOP CHECKPOINT ***
    A supervisor must approve this Refund request request:
    Draft response: Thank you for contacting Billing Support. Based on our records:
- SUBSCRIPTION PLANS
- REFUND POLICY
Customers may request a refund within 30 days of purchase if they are not satisfied with the service.
Annual subscription refunds are prorated based on unused months and require supervisor approval before processing.
Refunds for monthly subscriptions are processed within 5-7 business days after approval.
No refunds are issued after 30 days except in cases of proven billing errors.
- SUBSCRIPTION CANCELLATION POLICY
Customers can cancel their subscription at any time from the account settings page.
Cancellation requests for annual plans must be reviewed by a supervisor to confirm eligibility for partial refunds.
Once cancelled, customer data is retained for 90 days before permanent deletion.

This request has been flagged for supervisor approval before we can proceed.
    -> Supervisor decision: APPROVE (Approved per 30-day refund policy.)

    [Intent Classification] 'I need a refund for my annual subscription.' -> intent=Billing, department=Billing Support
    [Billing Support] RAG retrieved 3 chunk(s) from ['company_policy.txt', 'pricing_guide.txt']. High-risk=True (Refund request).
    [Human Approval] Supervisor decision: APPROVE — comment: Approved per 30-day refund policy.
    [Supervisor] Supervisor approved the high-risk request. Draft response validated and approved for sending.
    [Memory Save] Interaction persisted to memory.db (SQLite).

FINAL RESPONSE TO CUSTOMER:
   Thank you for contacting Billing Support. Based on our records:
- SUBSCRIPTION PLANS
- REFUND POLICY
Customers may request a refund within 30 days of purchase if they are not satisfied with the service.
Annual subscription refunds are prorated based on unused months and require supervisor approval before processing.
Refunds for monthly subscriptions are processed within 5-7 business days after approval.
No refunds are issued after 30 days except in cases of proven billing errors.
- SUBSCRIPTION CANCELLATION POLICY
Customers can cancel their subscription at any time from the account settings page.
Cancellation requests for annual plans must be reviewed by a supervisor to confirm eligibility for partial refunds.
Once cancelled, customer data is retained for 90 days before permanent deletion.

This request has been flagged for supervisor approval before we can proceed.
```

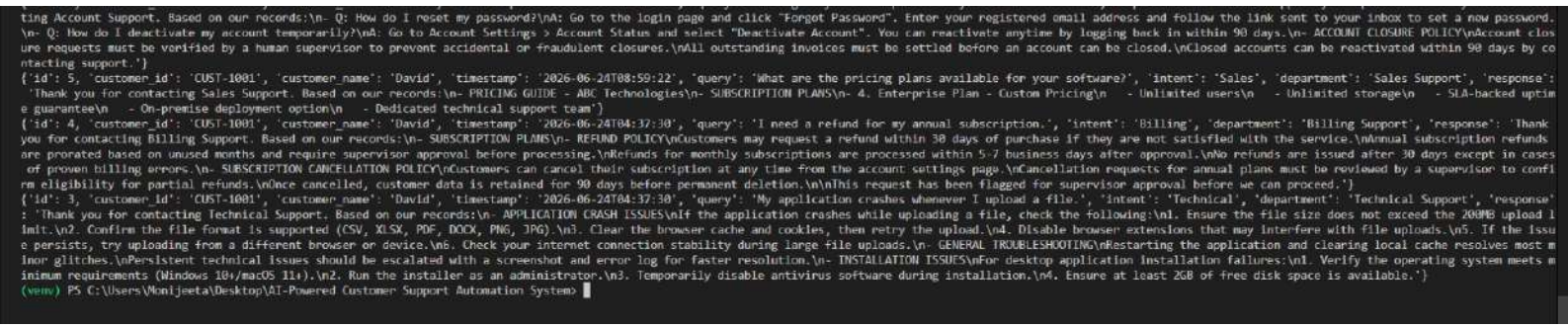
```
### QUERY 5 ###
================================================================================
CUSTOMER (David): What was my previous support issue?
================================================================================

    [Intent Classification] 'What was my previous support issue?' -> intent=Memory, department=Memory Recall
    [Memory Recall] Retrieved conversation history from SQLite memory.db
    [Memory Save] Skipped — this was a memory-recall query, not a new issue.

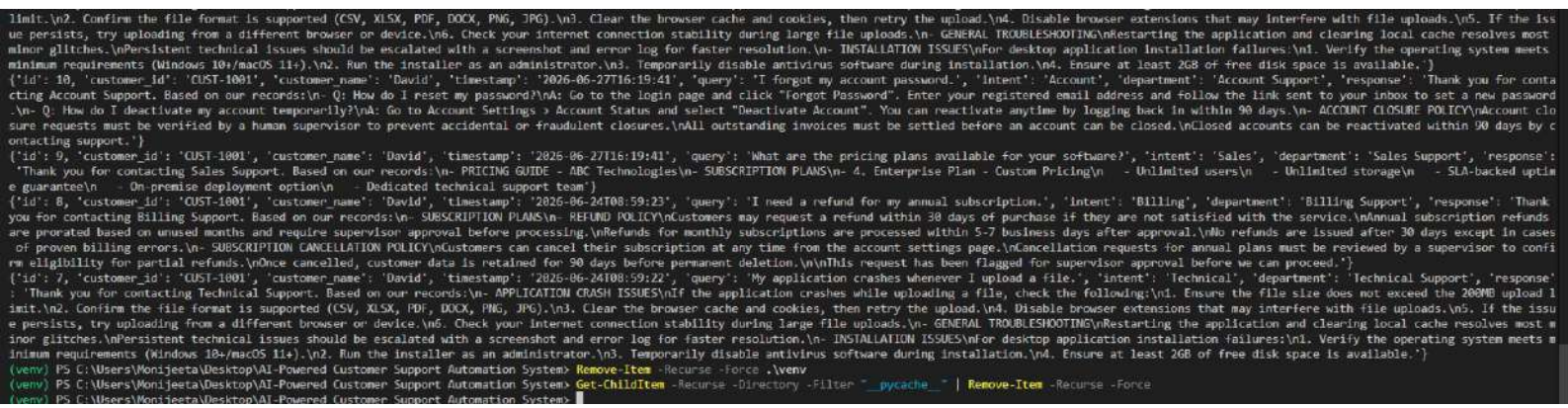
FINAL RESPONSE TO CUSTOMER:
   Hi David, your previous support request was: "I need a refund for my annual subscription." (handled by Billing Support on 2026-06-27T16:19:41). Our response at the time was: "Thank you for contacting Billing Support. Based on
our records:
- SUBSCRIPTION PLANS
- REFUND POLICY
Customers may request a refund within 30 days of purchase if they are not satisfied with the service.
Annual subscription refunds are prorated based on unused months and require supervisor approval before processing.
Refunds for monthly subscriptions are processed within 5-7 business days after approval.
No refunds are issued after 30 days except in cases of proven billing errors.
- SUBSCRIPTION CANCELLATION POLICY
Customers can cancel their subscription at any time from the account settings page.
Cancellation requests for annual plans must be reviewed by a supervisor to confirm eligibility for partial refunds.
Once cancelled, customer data is retained for 90 days before permanent deletion.

This request has been flagged for supervisor approval before we can proceed."
```

```
================================================================================
Full conversation history stored in memory.db for CUST-1001
================================================================================
{'id': 12, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'I need a refund for my annual subscription.', 'intent': 'Billing', 'department': 'Billing Support', 'response': 'Thank
 you for contacting Billing Support. Based on our records:\n- SUBSCRIPTION PLANS\n- REFUND POLICY\nCustomers may request a refund within 30 days of purchase if they are not satisfied with the service.\nAnnual subscription refunds
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irm eligibility for partial refunds.\nOnce cancelled, customer data is retained for 90 days before permanent deletion.\n\nThis request has been flagged for supervisor approval before we can proceed.'}
{'id': 11, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'My application crashes whenever I upload a file.', 'intent': 'Technical', 'department': 'Technical Support', 'response
': 'Thank you for contacting Technical Support. Based on our records:\n- APPLICATION CRASH ISSUES\nIf the application crashes while uploading a file, check the following:\n1. Ensure the file size does not exceed the 200MB upload
 limit.\n2. Confirm the file format is supported (CSV, XLSX, PDF, DOCX, PNG, JPG).\n3. Clear the browser cache and cookies, then retry the upload.\n4. Disable browser extensions that may interfere with file uploads.\n5. If the iss
ue persists, try uploading from a different browser or device.\n6. Check your internet connection stability during large file uploads.\n- GENERAL TROUBLESHOOTING\nRestarting the application and clearing local cache resolves most
 minor glitches.\nPersistent technical issues should be escalated with a screenshot and error log for faster resolution.\n- INSTALLATION ISSUES\nFor desktop application installation failures:\n1. Verify the operating system meets
 minimum requirements (Windows 10+/macOS 11+).\n2. Run the installer as an administrator.\n3. Temporarily disable antivirus software during installation.\n4. Ensure at least 2GB of free disk space is available.'}
{'id': 10, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'I forgot my account password.', 'intent': 'Account', 'department': 'Account Support', 'response': 'Thank you for conta
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.\n- Q: How do I deactivate my account temporarily?\nA: Go to Account Settings > Account Status and select "Deactivate Account". You can reactivate anytime by logging back in within 90 days.\n- ACCOUNT CLOSURE POLICY\nAccount clo
sure requests must be verified by a human supervisor to prevent accidental or fraudulent closures.\nAll outstanding invoices must be settled before an account can be closed.\nClosed accounts can be reactivated within 90 days by c
ontacting support.'}
{'id': 9, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'What are the pricing plans available for your software?', 'intent': 'Sales', 'department': 'Sales Support', 'response':
 'Thank you for contacting Sales Support. Based on our records:\n- PRICING GUIDE - ABC Technologies\n- SUBSCRIPTION PLANS\n- 4. Enterprise Plan - Custom Pricing\n   - Unlimited users\n   - Unlimited storage\n   - SLA-backed uptim
e guarantee\n   - On-premise deployment option\n   - Dedicated technical support team'}
{'id': 8, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T08:59:23', 'query': 'I need a refund for my annual subscription.', 'intent': 'Billing', 'department': 'Billing Support', 'response': 'Thank
you for contacting Billing Support. Based on our records:\n- SUBSCRIPTION PLANS\n- REFUND POLICY\nCustomers may request a refund within 30 days of purchase if they are not satisfied with the service.\nAnnual subscription refunds
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rm eligibility for partial refunds.\nOnce cancelled, customer data is retained for 90 days before permanent deletion.\n\nThis request has been flagged for supervisor approval before we can proceed.'}
{'id': 7, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T08:59:22', 'query': 'My application crashes whenever I upload a file.', 'intent': 'Technical', 'department': 'Technical Support', 'response'
: 'Thank you for contacting Technical Support. Based on our records:\n- APPLICATION CRASH ISSUES\nIf the application crashes while uploading a file, check the following:\n1. Ensure the file size does not exceed the 200MB upload l
imit.\n2. Confirm the file format is supported (CSV, XLSX, PDF, DOCX, PNG, JPG).\n3. Clear the browser cache and cookies, then retry the upload.\n4. Disable browser extensions that may interfere with file uploads.\n5. If the issu
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inor glitches.\nPersistent technical issues should be escalated with a screenshot and error log for faster resolution.\n- INSTALLATION ISSUES\nFor desktop application installation failures:\n1. Verify the operating system meets m
inimum requirements (Windows 10+/macOS 11+).\n2. Run the installer as an administrator.\n3. Temporarily disable antivirus software during installation.\n4. Ensure at least 2GB of free disk space is available.'}
{'id': 6, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T08:59:22', 'query': 'I forgot my account password.', 'intent': 'Account', 'department': 'Account Support', 'response': 'Thank you for contac
ting Account Support. Based on our records:\n- Q: How do I reset my password?\nA: Go to the login page and click "Forgot Password". Enter your registered email address and follow the link sent to your inbox to set a new password.
\n- Q: How do I deactivate my account temporarily?\nA: Go to Account Settings > Account Status and select "Deactivate Account". You can reactivate anytime by logging back in within 90 days.\n- ACCOUNT CLOSURE POLICY\nAccount clos
ure requests must be verified by a human supervisor to prevent accidental or fraudulent closures.\nAll outstanding invoices must be settled before an account can be closed.\nClosed accounts can be reactivated within 90 days by co
ntacting support.'}
{'id': 5, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T08:59:22', 'query': 'What are the pricing plans available for your software?', 'intent': 'Sales', 'department': 'Sales Support', 'response':
 'Thank you for contacting Sales Support. Based on our records:\n- PRICING GUIDE - ABC Technologies\n- SUBSCRIPTION PLANS\n- 4. Enterprise Plan - Custom Pricing\n   - Unlimited users\n   - Unlimited storage\n   - SLA-backed uptim
e guarantee\n   - On-premise deployment option\n   - Dedicated technical support team'}
{'id': 4, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T04:37:30', 'query': 'I need a refund for my annual subscription.', 'intent': 'Billing', 'department': 'Billing Support', 'response': 'Thank
you for contacting Billing Support. Based on our records:\n- SUBSCRIPTION PLANS\n- REFUND POLICY\nCustomers may request a refund within 30 days of purchase if they are not satisfied with the service.\nAnnual subscription refunds
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rm eligibility for partial refunds.\nOnce cancelled, customer data is retained for 90 days before permanent deletion.\n\nThis request has been flagged for supervisor approval before we can proceed.'}
```


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ntacting support.'}
{'id': 5, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T08:59:22', 'query': 'What are the pricing plans available for your software?', 'intent': 'Sales', 'department': 'Sales Support', 'response':
'Thank you for contacting Sales Support. Based on our records:\n- PRICING GUIDE - ABC Technologies\n- SUBSCRIPTION PLANS\n- 4. Enterprise Plan - Custom Pricing\n - Unlimited users\n - Unlimited storage\n - SLA-backed uptim
e guarantee\n - On-premise deployment option\n - Dedicated technical support team'}
{'id': 4, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T04:37:30', 'query': 'I need a refund for my annual subscription.', 'intent': 'Billing', 'department': 'Billing Support', 'response': 'Thank
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{'id': 3, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T04:37:30', 'query': 'My application crashes whenever I upload a file.', 'intent': 'Technical', 'department': 'Technical Support', 'response'
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(venv) PS C:\Users\Manjeeta\Desktop\AI-Powered Customer Support Automation System>

```
--------------------------------------------------------------------------------
Full conversation history stored in memory.db for CUST-1001
--------------------------------------------------------------------------------
{'id': 16, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:53:31', 'query': 'I need a refund for my annual subscription.', 'intent': 'Billing', 'department': 'Billing Support', 'response': 'Thank
 you for contacting Billing Support. Based on our records:\n- SUBSCRIPTION PLANS\n- REFUND POLICY\nCustomers may request a refund within 30 days of purchase if they are not satisfied with the service.\nAnnual subscription refunds
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irm eligibility for partial refunds.\nOnce cancelled, customer data is retained for 90 days before permanent deletion.\n\nThis request has been flagged for supervisor approval before we can proceed.'}
{'id': 15, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:53:31', 'query': 'My application crashes whenever I upload a file.', 'intent': 'Technical', 'department': 'Technical Support', 'response
': 'Thank you for contacting Technical Support. Based on our records:\n- APPLICATION CRASH ISSUES\nIf the application crashes while uploading a file, check the following:\n1. Ensure the file size does not exceed the 200MB upload
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{'id': 14, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:53:31', 'query': 'I forgot my account password.', 'intent': 'Account', 'department': 'Account Support', 'response': 'Thank you for conta
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sure requests must be verified by a human supervisor to prevent accidental or fraudulent closures.\nAll outstanding invoices must be settled before an account can be closed.\nClosed accounts can be reactivated within 90 days by c
ontacting support.'}
{'id': 13, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:53:31', 'query': 'What are the pricing plans available for your software?', 'intent': 'Sales', 'department': 'Sales Support', 'response'
: 'Thank you for contacting Sales Support. Based on our records:\n- PRICING GUIDE - ABC Technologies\n- SUBSCRIPTION PLANS\n- 4. Enterprise Plan - Custom Pricing\n   - Unlimited users\n   - Unlimited storage\n   - SLA-backed upti
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{'id': 12, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'I need a refund for my annual subscription.', 'intent': 'Billing', 'department': 'Billing Support', 'response': 'Thank
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{'id': 11, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'My application crashes whenever I upload a file.', 'intent': 'Technical', 'department': 'Technical Support', 'response
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{'id': 10, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'I forgot my account password.', 'intent': 'Account', 'department': 'Account Support', 'response': 'Thank you for conta
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{'id': 9, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'What are the pricing plans available for your software?', 'intent': 'Sales', 'department': 'Sales Support', 'response':
 'Thank you for contacting Sales Support. Based on our records:\n- PRICING GUIDE - ABC Technologies\n- SUBSCRIPTION PLANS\n- 4. Enterprise Plan - Custom Pricing\n   - Unlimited users\n   - Unlimited storage\n   - SLA-backed uptim
e guarantee\n   - On-premise deployment option\n   - Dedicated technical support team'}
{'id': 8, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T08:59:23', 'query': 'I need a refund for my annual subscription.', 'intent': 'Billing', 'department': 'Billing Support', 'response': 'Thank
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 of proven billing errors.\n- SUBSCRIPTION CANCELLATION POLICY\nCustomers can cancel their subscription at any time from the account settings page.\nCancellation requests for annual plans must be reviewed by a supervisor to confi
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limit.\n2. Confirm the file format is supported (CSV, XLSX, PDF, DOCX, PNG, JPG).\n3. Clear the browser cache and cookies, then retry the upload.\n4. Disable browser extensions that may interfere with file uploads.\n5. If the issue persists, try uploading from a different browser or device.\n6. Check your internet connection stability during large file uploads.\n- GENERAL TROUBLESHOOTING\nRestarting the application and clearing local cache resolves most minor glitches.\nPersistent technical issues should be escalated with a screenshot and error log for faster resolution.\n- INSTALLATION ISSUES\nFor desktop application installation failures:\n1. Verify the operating system meets minimum requirements (Windows 10+/macOS 11+).\n2. Run the installer as an administrator.\n3. Temporarily disable antivirus software during installation.\n4. Ensure at least 2GB of free disk space is available.'}
{'id': 10, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'I forgot my account password.', 'intent': 'Account', 'department': 'Account Support', 'response': 'Thank you for contacting Account Support. Based on our records:\n- Q: How do I reset my password?\nA: Go to the login page and click "Forgot Password". Enter your registered email address and follow the link sent to your inbox to set a new password.\n- Q: How do I deactivate my account temporarily?\nA: Go to Account Settings > Account Status and select "Deactivate Account". You can reactivate anytime by logging back in within 90 days.\n- ACCOUNT CLOSURE POLICY\nAccount closure requests must be verified by a human supervisor to prevent accidental or fraudulent closures.\nAll outstanding invoices must be settled before an account can be closed.\nClosed accounts can be reactivated within 90 days by contacting support.'}
{'id': 9, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-27T16:19:41', 'query': 'What are the pricing plans available for your software?', 'intent': 'Sales', 'department': 'Sales Support', 'response': 'Thank you for contacting Sales Support. Based on our records:\n- PRICING GUIDE - ABC Technologies\n- SUBSCRIPTION PLANS\n- 4. Enterprise Plan - Custom Pricing\n - Unlimited users\n - Unlimited storage\n - SLA-backed uptime guarantee\n - On-premise deployment option\n - Dedicated technical support team'}
{'id': 8, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T08:59:23', 'query': 'I need a refund for my annual subscription.', 'intent': 'Billing', 'department': 'Billing Support', 'response': 'Thank you for contacting Billing Support. Based on our records:\n- SUBSCRIPTION PLANS\n- REFUND POLICY\nCustomers may request a refund within 30 days of purchase if they are not satisfied with the service.\nAnnual subscription refunds are prorated based on unused months and require supervisor approval before processing.\nRefunds for monthly subscriptions are processed within 5-7 business days after approval.\nNo refunds are issued after 30 days except in cases of proven billing errors.\n- SUBSCRIPTION CANCELLATION POLICY\nCustomers can cancel their subscription at any time from the account settings page.\nCancellation requests for annual plans must be reviewed by a supervisor to confirm eligibility for partial refunds.\nOnce cancelled, customer data is retained for 90 days before permanent deletion.\n\nThis request has been flagged for supervisor approval before we can proceed.'}
{'id': 7, 'customer_id': 'CUST-1001', 'customer_name': 'David', 'timestamp': '2026-06-24T08:59:22', 'query': 'My application crashes whenever I upload a file.', 'intent': 'Technical', 'department': 'Technical Support', 'response': 'Thank you for contacting Technical Support. Based on our records:\n- APPLICATION CRASH ISSUES\nIf the application crashes while uploading a file, check the following:\n1. Ensure the file size does not exceed the 200MB upload limit.\n2. Confirm the file format is supported (CSV, XLSX, PDF, DOCX, PNG, JPG).\n3. Clear the browser cache and cookies, then retry the upload.\n4. Disable browser extensions that may interfere with file uploads.\n5. If the issue persists, try uploading from a different browser or device.\n6. Check your internet connection stability during large file uploads.\n- GENERAL TROUBLESHOOTING\nRestarting the application and clearing local cache resolves most minor glitches.\nPersistent technical issues should be escalated with a screenshot and error log for faster resolution.\n- INSTALLATION ISSUES\nFor desktop application installation failures:\n1. Verify the operating system meets minimum requirements (Windows 10+/macOS 11+).\n2. Run the installer as an administrator.\n3. Temporarily disable antivirus software during installation.\n4. Ensure at least 2GB of free disk space is available.'}
(venv) PS C:\Users\Monijeeta\Desktop\AI-Powered Customer Support Automation System> Remove-Item -Recurse -Force .\venv
(venv) PS C:\Users\Monijeeta\Desktop\AI-Powered Customer Support Automation System> Get-ChildItem -Recurse -Directory -Filter "__pycache__" | Remove-Item -Recurse -Force
(venv) PS C:\Users\Monijeeta\Desktop\AI-Powered Customer Support Automation System>